Schottky Barrier Rectifier, Trench-based

NRTS12100PFS, NRVTS12100PFS

This TO-277 trench Schottky rectifier provides fast switching performance in a compact thermally efficient package. The TO-277 package provides an excellent alternative to the DPAK, offering thermal performance nearly as good in a package occupying less than half the board space. Its low profile makes it a good option for flat panel display and other applications with limited vertical clearance. The device offers low leakage over temperature making it a good match for applications requiring low quiescent current.

Features

- Package Provides Capability of Inspection and Probe After Board Mounting
- Low Forward Voltage Drop
- 175 °C Operating Junction Temperature
- NRV Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable
- These Devices are Pb-Free, Halogen Free/BFR Free and are RoHS Compliant

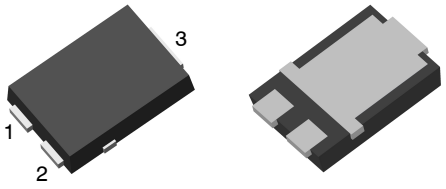
Mechanical Characteristics:

- Case: Epoxy, Molded
- Epoxy Meets Flammability Rating UL 94-0 @ 0.125 in.
- Lead Finish: 100% Matte Sn (Tin)
- Lead and Mounting Surface Temperature for Soldering Purposes: 260 °C Max. for 10 Seconds
- Device Meets MSL 1 Requirements

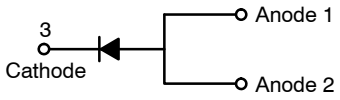
Applications

- Excellent Alternative to DPAK in Space-Constrained Automotive Applications
- Low Leakage for Higher Temperature Operation
- Output Rectification in Compact Portable Consumer Applications
- Freewheeling Diode used with Inductive Loads

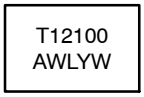
SCHOTTKY BARRIER RECTIFIER, 12 AMPERES 100 VOLTS



TO-277-3LD
CASE 340CZ



MARKING DIAGRAM



T12100	= Specific Device Code
A	= Assembly Location
Y	= Year
W	= Work Week
WL	= Wafer Lot

ORDERING INFORMATION

Device	Package	Shipping†
NRTS12100PFST3G	TO-277 (Pb-Free)	1500 / Tape & Reel
NRVTS12100PFST3G	TO-277 (Pb-Free)	1500 / Tape & Reel

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

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MAXIMUM RATINGS

Symbol	Rating	Value	Unit
V_{RRM} V_{RWM} V_R	Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	100	V
$I_{F(AV)}$	Average Rectified Forward Current (T_C = 160 °C)	12	A
I_{FRM}	Peak Repetitive Forward Current, (T_C = 157 °C, Square Wave, Duty = 0.5)	24	A
I_{FSM}	Non-Repetitive Peak Surge Current (Surge Applied at Rated Load Conditions Halfwave, Single Phase, 60 Hz)	130	A
T_{stg}	Storage Temperature Range	-65 to +175	°C
T_J	Operating Junction Temperature	–55 to +175	°C
	ESD Rating (Human Body Model)	3B	
	ESD Rating (Machine Model)	M4	

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS

Symbol	Characteristic	Max	Unit
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient (Assumes 600 mm^2, 1 oz. copper bond pad on a FR4 board)	69	°C/W
$R_{\theta JCT}$	Thermal Resistance, Junction-to-Case, Top (Assumes 600 mm^2, 1 oz. copper bond pad on a FR4 board)	59	°C/W
$R_{\theta JCB}$	Thermal Resistance, Junction-to-Case, Bottom (Assumes 600 mm^2, 1 oz. copper bond pad on a FR4 board)	1.92	°C/W

ELECTRICAL CHARACTERISTICS

Symbol	Characteristic	Typ	Max	Unit
v_F	Instantaneous Forward Voltage (Note 1) (i_F = 6 A, T_J = 25 °C) (i_F = 6 A, T_J = 125 °C) (i_F = 12 A, T_J = 25 °C) (i_F = 12 A, T_J = 125 °C)	 0.59 0.55 0.75 0.67	 – – 0.80 0.71	V
i_R	Instantaneous Reverse Current (Note 1) (Rated dc Voltage, T_J = 25 °C) (Rated dc Voltage, T_J = 125 °C)	 6 4.8	 120 53	 μA mA
C_J	Junction Capacitance (V_R = 1 V, T_J = 125 °C, 1 MHz)	930	–	pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse Test: Pulse Width = 300 μs, Duty Cycle ≤ 2.0%.

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TYPICAL CHARACTERISTICS

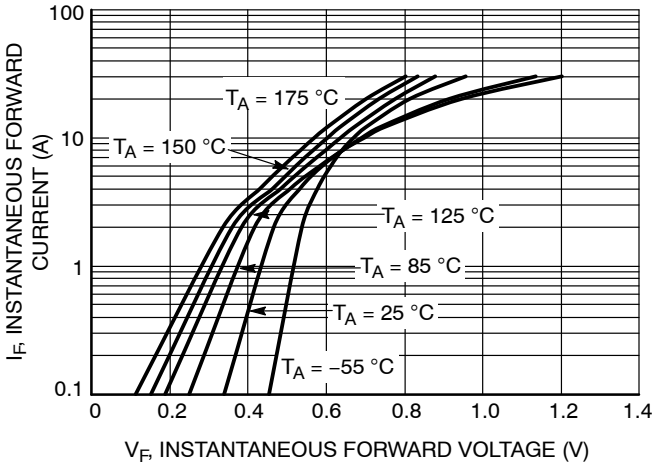


Figure 1. Typical Instantaneous Forward Characteristics

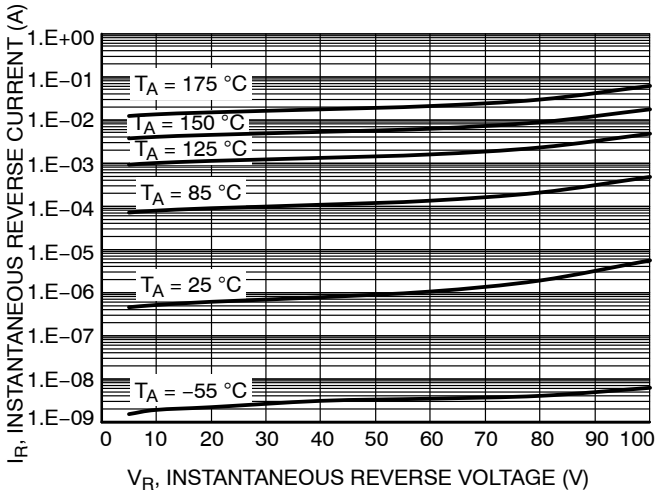


Figure 2. Typical Reverse Characteristics

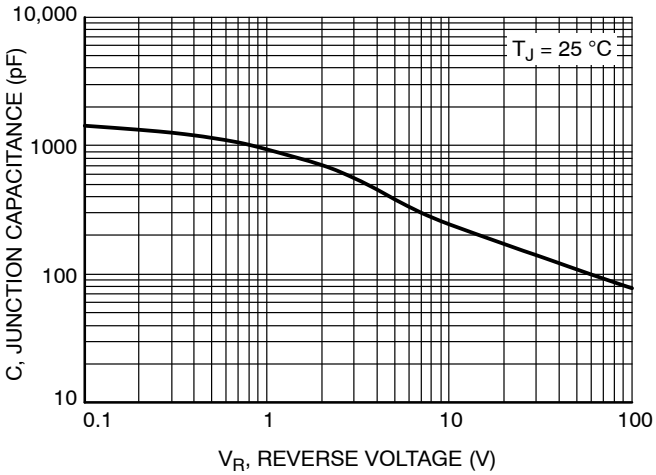


Figure 3. Typical Junction Capacitance

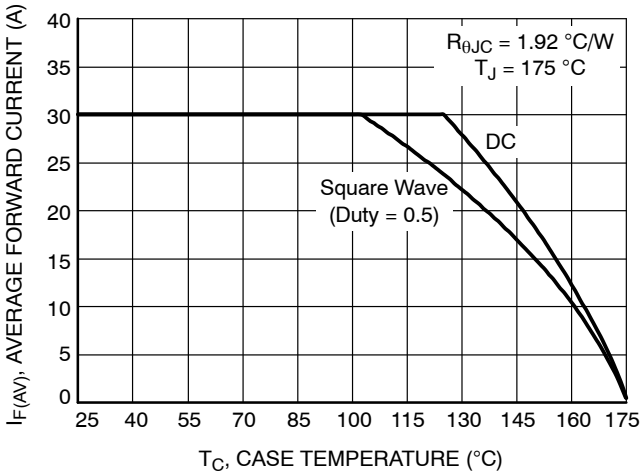


Figure 4. Current Derating

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TYPICAL CHARACTERISTICS (continued)

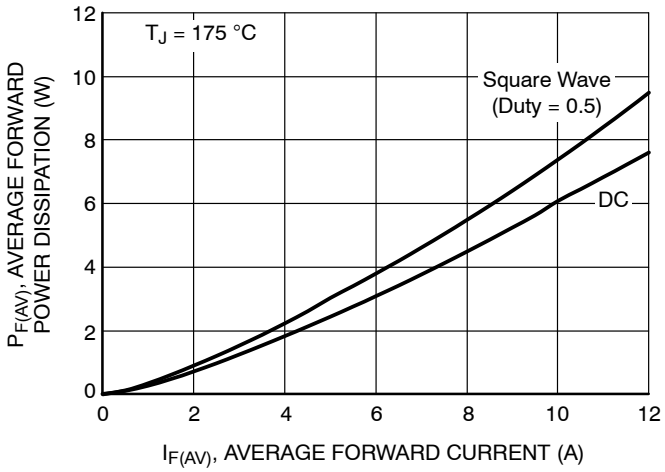


Figure 5. Forward Power Dissipation

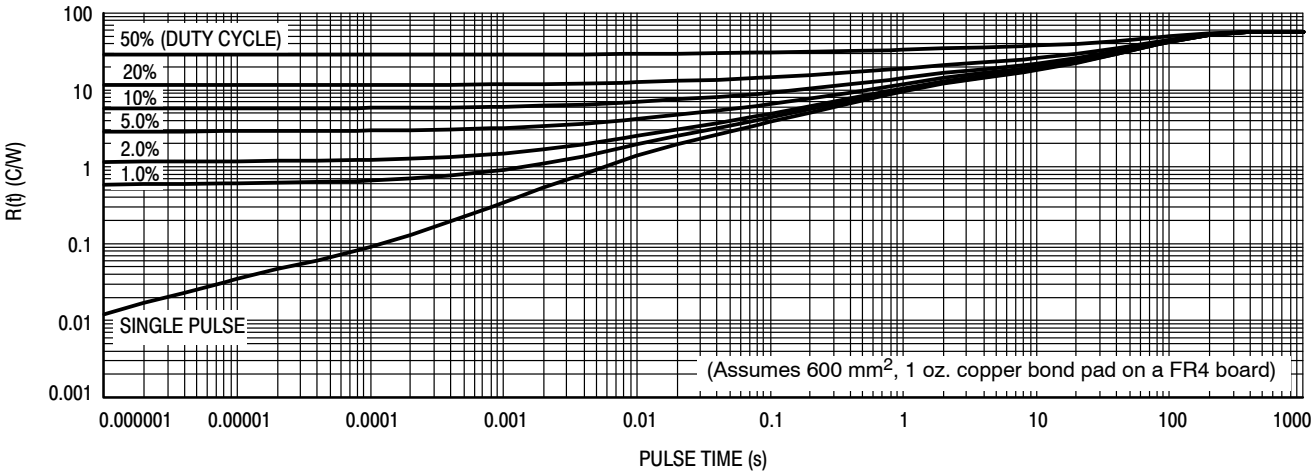


Figure 6. Typical Thermal Characteristics, Junction-to-Ambient

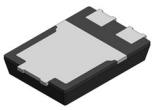
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REVISION HISTORY

Revision	Description of Changes	Date
1	Rebranded the Data Sheet to **onsemi** format.	7/9/2025

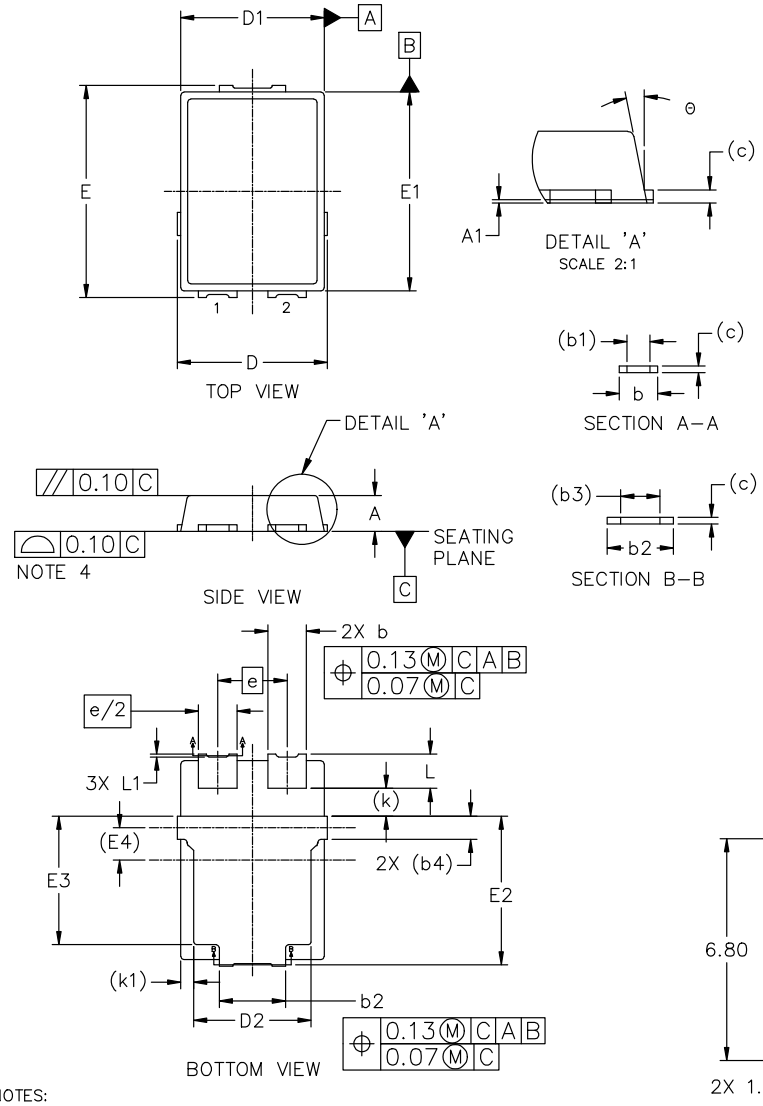
This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

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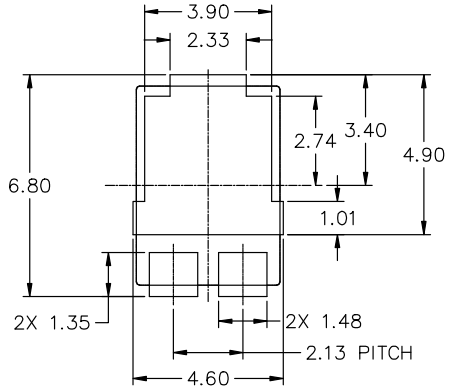


TO−277−3 4.40x6.10x1.10, 2.13P
CASE 340CZ
ISSUE B

DATE 04 MAY 2026



MILLIMETERS			
DIM	MIN	NOM	MAX
A	1.00	1.10	1.20
A1	−−−	0.01	0.05
b	1.13	1.18	1.28
b1		0.70 REF	
b2	1.98	2.03	2.13
b3		1.20 REF	
b4		0.71 REF	
c		0.20 REF	
D	4.45	4.60	4.75
D1	4.35	4.40	4.45
D2	3.50	3.60	3.70
E	6.35	6.50	6.65
E1	6.05	6.10	6.15
E2	4.50	4.60	4.70
E3	3.84	3.94	4.04
E4		0.98 REF	
e		2.13 BSC	
k		0.85 REF	
k1		0.40 REF	
L	0.90	1.05	1.20
L1	0.02	−−−	−−−
Θ	−−−	−−−	12°



RECOMMENDED MOUNTING FOOTPRINT

*For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERM/D.

NOTES:
1. DIMENSIONING AND TOLERANCING AS PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. DIMENSION b, b1, b2, b3, b4 and c TO BE MEASURED ON FLAT SECTION OF THE LEAD, BETWEEN 0.13 AND 0.25mm FROM LEAD TIP.
4. COPLANARITY APPLIES TO THE EXPOSED PAD AS WELL AS THE TERMINALS.
5. POSITIONAL TOLERANCE APPLIES TO THE TERMINALS AND EXPOSED PAD.
6. A1 IS DEFINED AS THE VERTICAL DISTANCE FROM THE SEATING PLANE TO THE LOWEST POINT ON THE PACKAGE BODY.
7. DIMENSION D1 AND E1 TO BE DETERMINED AT DATUM PLANE C.

GENERIC MARKING DIAGRAM*



XXXXXX = Specific Device Code
A = Assembly Location
Y = Year
W = Work Week
WL = Wafer Lot

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot " ▪", may or may not be present. Some products may not follow the Generic Marking.

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DESCRIPTION:	**TO−277−3 4.40x6.10x1.10, 2.13P**	**PAGE 1 OF 1**

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